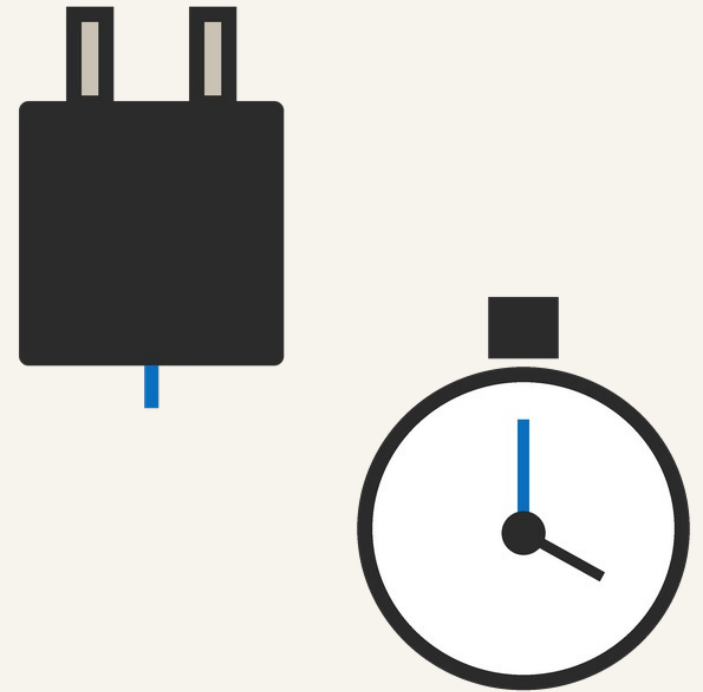


Monitoring & Measuring AC

A supply that never stops changing.

Higher Physics



Starter

2009 Higher Physics

Scan the code and answer these:

Q9

Q24

Section A · resistors, internal r
Show your working, not just the letter.



**Scan for the paper
Marking scheme**

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PART 1

What is alternating current?

Direct current flows one way; alternating current keeps changing its mind — and how fast tells you its frequency.

By the end of this part...

- 1 I can state what alternating current is

AC vs DC

DIRECT CURRENT (DC)

- Flows in ONE direction only
- Constant, or constant-ish, value
- e.g. a battery

ALTERNATING CURRENT (AC)

- Changes DIRECTION with time
- Instantaneous VALUE also changes with time
- e.g. the mains supply

The definition

- Alternating current is a current which changes direction AND changes its instantaneous value with time.
- BOTH parts of that definition are needed for full marks – "changes direction" alone is not enough.

Watch Out – the definition of AC

THE MISTAKE

- Describing AC as only changing direction
- Giving a graph description, not the definition

THE CORRECT ANSWER

- The definition needs BOTH parts: direction AND value.
- Learn the exact wording, not just a sketch.

Your turn — what is AC?

SHOW ME BOARDS

- 1 State the definition of alternating current.
- 2 State one everyday example of an AC supply.
- 3 State one everyday example of a DC supply.

TRY THIS – Answers

- 1 A current which changes direction and instantaneous value with time.
- 2 The mains supply.
- 3 A battery or cell.

Starter

2011 Higher Physics

Scan the code and answer these:



Section A · internal resistance

Show your working, not just the letter.



Scan for the paper

No marking scheme

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PART 2

Peak and rms values

The rms value is the steady DC current that would give the same heating effect as the real, ever-changing AC.

By the end of this part...

- 1 I can use $V_{\text{rms}} = V_{\text{peak}}/\sqrt{2}$ and $I_{\text{rms}} = I_{\text{peak}}/\sqrt{2}$ to solve problems

Why rms?

- An AC voltage is CONSTANTLY changing, so quoting its peak value overstates its real heating/power effect.
- The r.m.s. (root mean square) value is the equivalent STEADY DC value that delivers the SAME power.
- This is why a "230 V" mains supply means 230 V r.m.s. — its peak is actually higher.

Peak and rms relationships

$$V_{rms} = V_{peak} / \sqrt{2}$$

V_{rms}

r.m.s. voltage (V)

V_{peak}

peak voltage (V)

The same relationship links I_{rms} and I_{peak} .

Worked example – r.m.s. voltage

A supply has a peak voltage of 8.0 V. Find its r.m.s. voltage.

$$V_{\text{peak}} = 8.0 \text{ V}$$

$$V_{\text{rms}} = ?$$

$$V_{\text{rms}} = V_{\text{peak}} / \sqrt{2}$$

$$V_{\text{rms}} = 8.0 / \sqrt{2}$$

$$\underline{V_{\text{rms}} = 5.7 \text{ V}}$$

Your turn — peak and rms

SHOW ME BOARDS

- 1 A supply has a peak current of 4.0 A . Find its r.m.s. current.
- 2 The UK mains supply is 230 V r.m.s. Find its peak voltage.
- 3 State what "r.m.s." stands for.

TRY THIS – Answers

- 1 2.8 A.
- 2 325 V.
- 3 Root mean square.

Starter

2008 Higher Physics

Scan the code and answer these:

Q9

Section A · e.m.f. definition

Show your working, not just the letter.



**Scan for the paper
Marking scheme**

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PART 3

Frequency and period

How many complete cycles happen each second — one relationship, read either way round.

By the end of this part...

- 1 I can use $f = 1/T$ to solve problems involving frequency and period

$$\underline{f = 1/T}$$

$$f = 1/T$$

f

frequency (Hz)

T

period (s)

T is the time for ONE complete cycle – check the trace shows a full cycle before you measure it.

Worked example – frequency

An AC supply has a period of 0.020 s. Find its frequency.

$$T = 0.020 \text{ s}$$

$$f = ?$$

$$f = 1/T$$

$$f = 1 / 0.020$$

$$\underline{f = 50 \text{ Hz}}$$

Your turn – frequency and period

SHOW ME BOARDS

- 1 An AC supply has a frequency of 100 Hz. Find its period.
- 2 State the frequency of the UK mains supply.
- 3 State what T represents in $f = 1/T$.

TRY THIS – Answers

- 1 0.010 s.
- 2 50 Hz.
- 3 The period – the time for one complete cycle.

Starter

2012 Revised Higher Physics

Scan the code and answer these:

Q17

Q18

Section A · resistors, internal r
Show your working, not just the letter.



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PART 4

Reading the oscilloscope

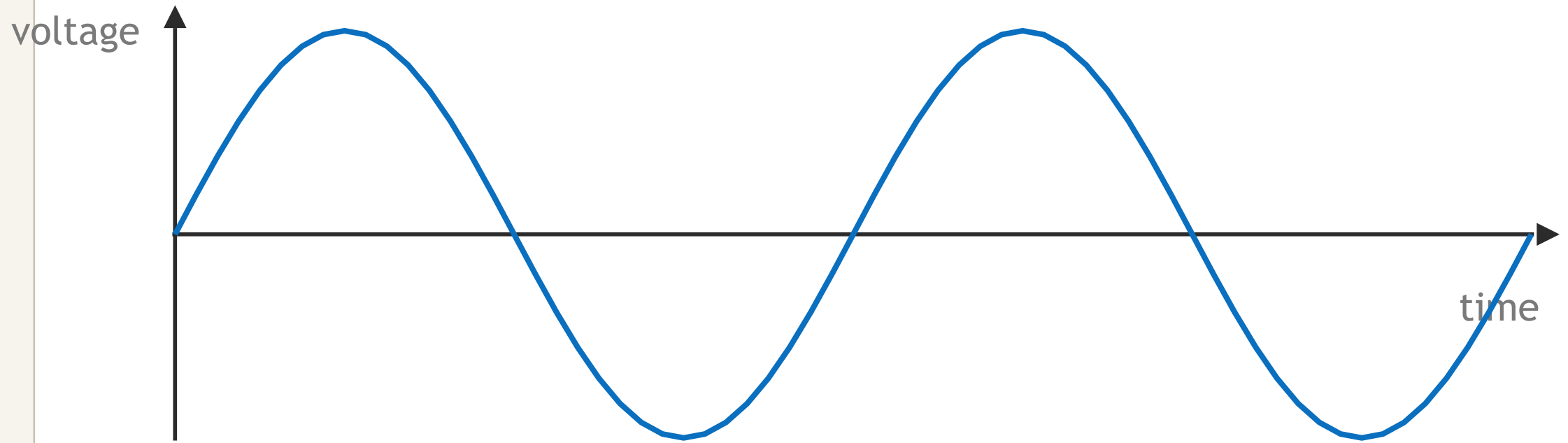
An oscilloscope turns two dial settings into a real peak voltage and a real frequency — read the grid, not just the shape.

By the end of this part...

1

I can determine frequency, peak and r.m.s. values from an oscilloscope trace

The trace



Two settings, two readings

- Y-gain (volts/division) $\times$ the peak height in divisions = V_{peak} .
- Timebase (seconds/division) $\times$ the divisions for ONE cycle = T — then $f = 1/T$.
- Count divisions carefully — a peak-to-peak height is TWICE the peak value.

Worked example – reading an oscilloscope trace

A trace has a peak height of 3.0 divisions with Y-gain 2.0 V/div, and one cycle spans 4.0 divisions with timebase 5.0 ms/div. Find V_{peak} , T , f and V_{rms} .

$$\text{Y-gain} = 2.0 \text{ V/div}$$

$$3.0 \text{ div peak}$$

$$\text{timebase} = 5.0 \text{ ms/div}$$

$$4.0 \text{ div/cycle}$$

$$V_{\text{peak}} = 3.0 \times 2.0 = 6.0 \text{ V}$$

$$T = 4.0 \times 5.0 \text{ ms} = 0.020 \text{ s}$$

$$f = 1/T = 1 / 0.020 = 50 \text{ Hz}$$

$$\underline{V_{\text{rms}} = V_{\text{peak}} / \sqrt{2} = 6.0 / \sqrt{2} = 4.2 \text{ V}}$$

Watch Out — oscilloscope readings

THE MISTAKE

- Reading peak-to-peak height as the peak value
- Forgetting to show where the period value came from

THE CORRECT ANSWER

- Peak-to-peak is TWICE V_{peak} — halve it first.
- Always show the division count and the setting used.

Your turn – the oscilloscope

SHOW ME BOARDS

- 1 A trace has a peak height of 4.0 divisions with Y-gain 1.5 V/div. Find V_{peak} .
- 2 One cycle spans 5.0 divisions with timebase 4.0 ms/div. Find the period and frequency.
- 3 State why peak-to-peak height must be halved before use.

TRY THIS – Answers

- 1 6.0 V.
- 2 $T = 0.020 \text{ s}$; $f = 50 \text{ Hz}$.
- 3 Peak-to-peak spans both the positive and negative peak, so it is twice the peak value.

Starter

2015 Revised Higher Physics

Scan the code and answer these:

Q13

Q30

Section A · uncertainty, internal r
Show your working, not just the letter.



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Marking scheme**

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PART 5

Bringing it together

Every AC question is one, two or all three of these relationships, applied to whatever the trace or the text gives you.

By the end of this part...

1

I can combine peak/rms and frequency/period relationships to solve multi-step AC problems



Monitoring and measuring AC – summary

AC changes direction AND instantaneous value with time.

$$V_{rms} = V_{peak} / \sqrt{2}$$

$$I_{rms} = I_{peak} / \sqrt{2}$$

$$f = 1/T$$

Worked example – the UK mains supply

The UK mains supply is 230 V r.m.s. with frequency 50 Hz.
Find the peak voltage and the period.

$$V_{\text{rms}} = 230 \text{ V}$$

$$f = 50 \text{ Hz}$$

$$V_{\text{peak}} = V_{\text{rms}} \sqrt{2}$$

$$V_{\text{peak}} = 230 \times \sqrt{2} = 325 \text{ V}$$

$$T = 1/f$$

$$\underline{T = 1 / 50 = 0.020 \text{ s}}$$

Checkpoint

- 1 State the definition of alternating current.
- 2 Use $V_{\text{rms}} = V_{\text{peak}} / \sqrt{2}$ to find V_{rms} given V_{peak} .
- 3 State the relationship between frequency and period.
- 4 State how V_{peak} and T are found from an oscilloscope trace.
- 5 Find the peak voltage of a 230 V r.m.s. supply.

Exam Practice

2025 Q11 – trace amplitude when a switch is closed.

2024 Q13 – the definition of AC, and rms voltage from a trace.

2023 Q5 – period of an AC supply from flashing LEDs.

Past papers, Section A

LOOKING AHEAD

Next: Capacitors

A component that stores charge instead of resisting it — and behaves completely differently on AC and DC.

Plenary

- 1 State the definition of alternating current.
- 2 State the relationship between rms and peak values.
- 3 State the relationship between frequency and period.
- 4 State how V_{peak} is found from an oscilloscope trace.
- 5 Find V_{rms} for a supply with $V_{\text{peak}} = 10.0 \text{ V}$.

- 1 Explain why household appliances are rated using r.m.s. values, not peak values.
- 2 A trace's timebase is halved. Explain what happens to the number of cycles shown on the screen.

- 1 Derive the peak voltage of the UK mains supply from its r.m.s. value.
- 2 Explain, using r.m.s., why a 230 V AC heater and a 230 V DC heater of the same resistance produce the same power.

Extension – EXTRA

EXTRA PRACTICE

- 1 Research the frequency of the mains supply in a country outside Europe and compare it to the UK's 50 Hz.